

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. Examination, 2024
CHE – 1104, Chemistry

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Write the Postulate of Rutherford atom model with limitations. 5
 b) Explain the dual nature of an electron. 3
 c) Why the last electron of K occupies the 4s orbital rather than 3d orbital? 3
 d) State and discuss in brief Heisenberg's uncertainty principle. 3
2. a) What are the s, p, d, and f-block elements? "All transition elements are d-block elements, but not all d-block elements are transition elements"- Explain this distinction. 2+3
 b) What is electronegativity? Why ionic radius of K^+ is smaller than that of Cl^- ? 4
 c) What are inert gases? Explain the causes why they are inert. 1+2
 d) Write down the electronic configurations of the following- 2
 Al^{+3}, S^{-2}
3. a) What do you understand by a chemical bond? How is an ionic bond formed? Explain with an example. 1+3
 b) Why do some covalent compounds exhibit ionic character? Explain with a suitable example. 4
 c) 'Coordinate bond is basically a special type of covalent bond.' – Explain. 3
 d) AgF is soluble in water, but why are $AgCl$, $AgBr$, and AgI insoluble in water? 3
4. a) Derive Raoult's law of relative lowering of vapor pressure. 4
 b) Classify the following acid and base on different acid base concept, and also justify your answer: NH_3 , CaO , NH_4^+ and $AlCl_3$. 4
 c) What is enthalpy? Why it is constant for strong acid and strong base? 3
 d) Calculate the P^H value of 0.1 (M) CH_3COOH solution. ($K_a = 1.8 \times 10^{-5}$). 3
5. a) What is K_p and K_c ? How K_p and K_c are related? 4
 b) Explain how equilibrium constant changes with temperature for exothermic and endothermic reaction. 4
 c) State law of mass action. 2
 d) Find K_p and K_c for the reaction, $H_2 + I_2 = 2HI$, where the degree of dissociation is α . 4
6. a) What do you understand by reversible and irreversible reactions? Give an example of each. 3
 b) The half-life of a first order reaction is 15 min. How long will it take to convert 70% reactant into product? 2+2
 c) Derive the equation of equilibrium constant for the following reaction- 3
 $2HCl \rightleftharpoons H_2 + Cl_2$
 d) Explain temperature effect on a reaction by Arrhenius equation. 1+3
7. a) What is electrolysis? State Faraday's laws of electrolysis. 1+3
 b) What is an electrochemical cell? Write the cell reaction of the cell 1+2
 $Zn/ZnSO_4//CuSO_4/Cu$
 c) Define oxidation potential and reduction potential. 'Corrosion is a chemical process' - explain the statement. 2+2
 d) State and explain Hess's law of constant heat summation. 3